

1. Infection of the human eye is very rare due to the following?

- a. viral killing by phagocytosis
- b. bactericidal activity of secretory IgA in the presence of lysozyme**
- c. Bactericidal activity of IgG and interferons
- d. highly efficient NK cells of the eye tears
- e. All of the choices are correct

2. The following immune molecules are structurally related EXCEPT?

- a. CDI
- b. MHC class II**
- e. FCPR or Brambeli receptor
- d. MHC class 1

3. BCR does not?

- a. undergoe Somatic DNA recombination
- b. recognize hiddenconformational epitopes**
- e undergoe somatic hypermutation
- d. recognize native nondegradedproteins

4. The function of allelic exclusion is?

- a. increase the magnitude of antibodies
- b. increase the affinity of antibodies
- c. increase the specificity of antibodies**
- d. increase, the diversity of antibodies
- e. increase sensitivity of antibodies

5. The following promote development of B cells?

- a. IL-7 and bone marrow stroma are correct**
- b. IL-7
- c. IL-6
- d. stroma of bone marrow

6. Thymic positive selection of T cells means?

- a. T cells that recognize non self peptides are deleted
- b. T cells that fail to recognize self-MHC are deleted.**
- c. T cells that recognize self-peptides are deleted
- d. T cells that recognize non self MHC are deleted

7. We develop self-tolerance to molecules normally found outside the thymus due to?

- a. AIRE gene**
- b. MHC gene
- c. CD40 gene
- d. TCR gene

8. The following applies for human and not birds?

- a. somatic gene conversion is responsible for antibody diversity
- b. Somatic DNA recombination contributes mainly to 11 cell diversity**
- c. lack of lymph nodes
- d. presence of bursa at Fabricius

9. Thymectomy of a 60-years old man results in?

- a. little decrease in T cells**
- b. dramatic (highly) decrease in B cells
- c. dramatic decrease in T cells
- d. repetitive infections due to severe immune deficiency
- e. Death of the patient

10. Gamma delta (??) T Cells respond to?

- a. viral DNA or RNA
- b. peptides and MHC only
- c. Nonprotein phosphor antigens
- d. lipopolysaccharides

11. Which of the following statements is TRUE about interferons?

- a. Interferons interfere with viral replication inside host cells
- b. Interferons attack and lyse the fungi
- c. Interferons are glycolipids produced by healthy uninfected cells
- d. Interferons are found only in parasites
- e. Interferons are specific for individual bacteria

12. Immature T cells are considered?

- a. have neither TCR nor CD3
- b. have both CD4 and CD8 (Double positive)
- c. have either CD4 or CD8 (single positive)
- d. have neither CD4 nor CD8 (Double negative)

13. RAG-1 and RAG-2 enzymes mediate the recombination of?

- a. VDJ to CH
- b. V to D
- c. CDR1 to CDR2
- d. H to L
- e. VJ to C

14. The following sentence is false about immune response (i.r)?

- a. Latent Period is longer in primary than in secondary
- b. antibody affinity is lower in primary i.r than in secondary
- c. log phase is higher in primary i.r than in secondary
- d. decline phase is shorter in primary i.r than in secondary

15. The following applies for pro -T cells?

- a. have CD4 expression
- b. the gene of TCR-beta chain starts to rearrange in the cortex
- c. undergoes positive selection
- d. Have MHC class II on surface

16. Splenectomy of a 60 years old man results in?

- a. Little impact because other organs take over (compensate)
- b. An abnormal increase of T cells and a decrease in B cells
- c. repetitive infections due to severe immunodeficiency
- d. death of the patient due to the loss of B cells

17. The following is a characteristic of mature plasma cells?

- a. express membrane C.D10
- b. express membrane Ig
- c. Generated In the germinal centers
- d. express MHC

18. Beta 2- microglobulin belong to?

- a. HLA-DR
- b. HLA-DQ
- c. HLA-C
- d. HLA-DP
- e. HLA-DM

19. The following is incorrect about somatic hypermutation?

- a. occurs in variable regions CDR hotspots of Ig genes H and L chains
- b. occurs only within germinal centers
- c. increase both the variety and the affinity of Abs
- d. Activation induced cytidine deaminase play a role
- e. affect all B cells and the mutations are transmitted to offspring

20. The following sentence is false about the antigenicity?

- a. Lipids are rarely immunogenic
- b. Side chains of polysaccharides (blood groups, teichoic acid) cannot activate T cells
- c. Native helical DNA is not immunogenic
- d. Glycoproteins evoke T cell dependent immune response against the sugar and against the protein
- e. Polysaccharides can be antigenic and activate T cells

21. The following is not a characteristic of mature B cells?

- a. CD10 expression
- b. Membrane IgM and IgD
- c. Signal transduction molecules: Ig alpha and Ig beta
- d. Surface coreceptor molecules: CD19, 81 and 21

22. The following organ is responsible for B cell development of the fetus in the first 3 months of pregnancy?

- a. thymus
- b. liver
- c. Bone marrow
- d. spleen

23. The following is not involved in B-T cell immunological synapsis?

- a. B7
- b. ICOSL
- c. CD40
- d. IL-2

24. The following cell does not have surface HLA-DR?

- a. Thymic epithelial cells
- b. B cell
- c. Activated T cell
- d. neutrophils

25. The paracortical zone of lymph nodes is rich in?

- a. mast cells
- b. NK cells
- c. T and B cells
- d. B cells and macrophages
- e. T and dendritic cells

26. Features of T-independent immuno response include?

select one:

- a. give rise to memory coils.
- b. production of IgG and generate IgM
- c. occurs in people or animals that lack a functional thymus
- d. immunological synapsis

27. The following sentence is false about adjuvants?

- a. covalently linked to an Ag in order to increase the number epitopes
- b. Incomplete Freund adjuvant does not have Mycobacterium, tuberculosis
- c. Alum is a licensed adjuvants for use in human vaccine
- d. Are vehicles that are mixed with an Ag solution immune response
- e. Freund complete adjuvant consist of killed M tuberculosis suspended in oil and aqueous solution

28. The following cell does not belong to innate immunity?

- a. Neutrophil
- b. NK cells
- c. dendritic cells
- d. B lymphocytes
- e. Monocyte

29. Neutrophils were isolated from the blood of the following 4 mice groups in separate tubes. Which group of neutrophils possess the ability to kill all gram negative bacteria?

- a. Group 4 was injected with PBS
- b. Group 1 was injected with the lipoteichoic acid of gram positive bacteria
- c. All of the choices are correct
- d. group 2 was injected with the LPS of gram negative bacteria
- e. group 3 was injected with double stranded viral RNA

30. IgA in human milk?

- a. bind to neutrophils
- b. Has J-chain
- c. Has secretory component
- d. All choices are correct
- e. Is efficient agglutinating antibody against viruses

31. The following cell does not contain HLA-A?

- a. Macrophages
- b. T cell
- c. B cell
- d. Erythrocytes

32. In well-felix test, cross reaction occurs between antibodies against?

- a. Epstein barr virus and horse RBCs
- b. Rickettsia typhi and proteus vulgaris bacteria
- c. E coli antigen and human blood groups B
- d. Cytomegalovirus and human proteins
- e. pneumococcus polysaccharide and human blood groups A

33. The following genes belong to class I human MHC region?

- a. HLA-A
- b. HLA-DR
- c. I-E
- d. K, D

34. The following cell describe RIG-like receptors?

- a. are TLR
- b. they sense foreign dsDNA
- c. induce production of interferon alpha
- ci. cytoplasmic DNA helicase
- e. recognize unmethylated nucleotides

35. HLA-G has the following function?

- a. participate in the self -tolerance of T cells
- b. involved in the peptide exchange
- c. involved in the transportation of peptides to ER
- c. prevent the rejection of the fetus by the mother

36. The following is not a characteristic of immature B cells?

- a. interaction with foreign Ag results in anergy
- b. expression of IgM on cell surface
- c. expression of IgD on cell surface
- d. Deletion of self-reactive B cells

37. The following is not a characteristic of memory B cells?

- a. None proliferating
- b. Long-lived
- c. express IgD
- d. Express CD27

38. BCR does not perform the following?

- a. undergoes somatic hypermutation
- b. recognize native nondegraded three dimensional proteins
- c. recognize hidden conformational epitopes
- d. undergoes somatic DNA recombination

39. The following is not a characteristic of mature plasma cells?

- a. express no MHC
- b. express membrane Ig
- c. Generated in the germinal centers
- d. express no membrane CD10

40. The following events does not occur in the germinal centers of activated B cells?

Select one:

- a. somatic hypermutation
- b. Somatic DNA recombination
- c. formation of memory B cells
- d. class switching
- e. plasma cell formation

41. CD4 T cells do all of these EXCEPT?

- a. Adhesion
- b. signal transduction
- c. bind MHC class I on APCs cells
- d. Receptor for HIV

42. Gamma delta ($\gamma\delta$) T cells respond to?

select one:

- a. Nonprotein phosphoantigens
- b. lipopolysaccharide
- c. viral DNA or RNA
- d. peptides and MHC only

43. The following does not characterizes junctional diversity that contribute to antibody diversity?

- a. occurs exclusively in immature lymphocytes
- b. both RAG-1 and RAG-2 are involved
- c. insertion of none-coded nucleotides to the ends
- d. Insertion of palindromic nucleotides to both ends
- e. TdT adds or remove nucleotides at the exposed ends of the V, D or J genes before they are reunited .

44. Which of the following statements is TRUE about IgE?

- a. IgE can bind to macrophages
- b. IgE is abundant in saliva
- c. IgE has a significant role in bacteria infections
- d. IgE activates the complement cascade
- e. IgE bind strongly to mast cells and basophils

45. The following is not APC?

- a. Macrophages
- b. b cell
- c. Dendritic cell
- d. activated T cell

46. Like BCR, TCR?

- a. contain 3 hypervariable regions
- b. appears in secreted form
- c. undergoes class switching
- d. undergoes somatic hypermutation

47. The following is not a characteristic of pre-B cells?

- a. surrogate L chain
- b. expression of CD10, the early Marker
- c. expression of IgM on cell surface
- d. Allelic exclusion occurs

48. The following act as a chaperone for the newly synthesized MHC?

- a. Ii chain
- b. TAP
- c. HLA-DM
- d. CD3

49. The following applies for pre - T cells ?

- a . undergo allelic exclusion of TCR beta chain gene
- b. have CD4 or CD8 expression
- c . undergoes positive selection
- d . Have MHC class on surface

50. Entry of mature resting lymphocytes to Lymph nodes is mediated by the expression of :

- a. IL – 7
- b . selectin
- c. TCR
- d . IL – 12
- e . IL - 2

51. The following participate in intracellular activation of T cells ? Select one :

- a . phospholipase γ
- b . NF - κ B transcription factor .
- c. All of the choices are correct
- d . ZAP – 70
- e . ITAM

52. The mechanism that permits one B cell to synthesize surface IgM and IgD that recognize only one epitope ?

- a. affinity maturation
- b. differential RNA processing or alternative splicing .
- c. codominant expression
- d. class switching
- e. allelic exclusion

53. An antigen binding fragment (Fab) :

- a. produced by mercaptoethanol treatment
- b. produced by papain treatment
- c. Has no interchain disulfide bond
- d. contain no variable region
- e. Lacks light chain

54. Five mice group 1,2,3,4 and 5 were injected as follows. Which group is expected to produce the highest amount of antibodies :

- a. Group 4 with LPS
- b. Group 1 was injected with an olive oil
- c. Group 2 with human cholesterol
- d. Group 5 with fructose
- e. Group 3 with viral dsRNA

55. The following is primary lymphoid organ :

- a. Thymus
- b. Lymph nodes
- c. MALT tissue
- d. Tonsils
- e. Spleen

56. Natural Killer (NK) cells do NOT :

- a. Kill by phagocytosis of the target cell
- b. contain tumor necrosis factor (TNF)
- c. contain serine proteases
- d. Respond to interferon
- e. contain perforin and granzymes

57. The following costimulatory pairs participate in the immunological synapse of APC-T cells :

- a. B7 --- CD40
- b. IL-2 ---- IL-2 receptor
- c. ICOSL --- ICOS
- d. MHC --- TCR

58. The function of Plasmacytoid dendritic cells is? Select one :

- a. secretion of IFN α and β .
- b. induction of plasma cells
- c . inhibition of immune response
- d. induction of T cells

59. All the following are characteristics of both MHC class I and II molecules EXCEPT :

- A) They are expressed codominantly.
- B) They expressed constitutively on all nucleated cells
- C) They are polypeptides with domain structure.
- D) They are involved in presentation of antigen fragments to T cells.
- E) They are expressed on the surface membrane of B cells

60. MHC class I molecules are important for which of the following :

- A) binding to CD8 molecules on T cells
- B) presenting exogenous antigen (e.g., bacterial protein) to B cells
- C) presenting intact viral proteins to T cells
- D) binding to CD4 molecules on T cells
- E) binding to Ig on B cells.

61. Which of the following is incorrect concerning MHC class II molecules?

- A) B cells may express different allelic forms of MHC class molecules on their surface.
- B) MHC class II molecules are synthesized in the endoplasmic reticulum of APCs.
- C) Genetically different individuals express different MHC class II alleles.
- D) MHC class II molecules are associated with β 2-microglobulin on the cell surface.**
- E) A peptide that does not bind to an MHC class II molecule will not trigger a CD4+ T-cell response.

62. The earliest stages of B-cell differentiation:

- a. occur in the embryonic thymus
- b. require the presence of antigen
- c. involve rearrangement of κ -chain gene segments
- d. involve rearrangement of surrogate light-chain gene segments
- e. involve rearrangement of heavy-chain gene segments**

63. Immature B lymphocytes:

- A) have rearranged only D and J gene segments
- B) are progenitors of T as well as B lymphocytes
- C) express both IgM and IgD on their surfaces
- D) are at a stage of development where contact with antigen may lead to receptor editing and deletion**
- E) must go through the thymus to mature

64. Germinal centers found in lymph nodes and spleen:

- A) support the development of immature B and T cells
- B) function in the removal of damaged erythrocytes from the circulation
- C) act as the major source of stem cells and thus help to maintain hematopoiesis
- D) are sites where antigen-activated mature B cells proliferate and differentiate
- E) exclude T cells

65. Which of the following is expressed on the surface of the mature B lymphocyte?

- A) CD40
- B) MHC class II molecules
- C) CD32
- D) IgM and IgD
- E) all of the above

66. Which of the following would not be found on a memory B cell?

- A) Ig α and Ig β
- B) γ heavy chains
- C) ϵ heavy chains
- D) surrogate light chains
- E) κ light chains

67. Which of the following statements is incorrect?

- A) Antibodies in a secondary immune response generally have a higher affinity for antigen than antibodies formed in a primary response.
- B) Somatic hypermutation of variable-region genes may contribute to changes in antibody affinity observed during secondary responses.
- C) Synthesis of antibody in a primary response to a thymus-dependent antigen occurs predominantly in the blood.
- D) Isotype switching occurs in the presence of antigen.
- E) Predominantly IgM antibody is produced in the primary response

68. The following cell does not have toll-like receptors :

- a) Neutrophil
- b) Monocyte
- c) B lymphocytes
- d) Macrophages

69. The following is an antigen –presenting cell :

- a) Neutrophil
- b) Monocyte
- c) B lymphocytes
- d) Dendritic cells

70. Which of the following does not protect body surface :

- a) Mucus
- b) Gastric acid
- c) skin
- d) Salivary amylase
- e) Gut microflora

71. The following cell secrete antibodies :

- a) Neutrophil
- b) Monocyte
- c) B lymphocytes
- d) NK cells

72. Which of the following statements is TRUE of interferons?

- a. Interferons are glycolipids produced by healthy uninfected cell
- b. Interferons attack and lyse the fungi
- c. Interferons are specific for individual bacteria.
- d. Interferons are proteins against viruses

73. Which of the following statements is TRUE about lysozyme? Select one:

- a. Lysozyme is released by mast cells
- b. Lysozyme is a proteolytic enzyme
- c. Lysozyme splits peptidoglycan
- d. Lysozyme activates complement
- e. Lysozyme is a cytoplasmic organelle

74. Polymorphonuclear neutrophils attack bacteria? Select one:

- a. By phagocytosis
- b. By secreting complement
- c. By secreting granzymes and perforins
- d. By secreting interferon
- e. Exclusively by oxygen dependent mechanisms

75. The following cell represent the link between innate immunity and adaptive:

- a) Neutrophil
- b) Monocyte
- c) B lymphocytes
- d) Dendritic cells

76. Killer-cell inhibitory receptors (KIRs) expressed by NK cells bind to which of the following to prevent killing of normal cell:

- A) complement receptors
- B) MHC class I
- C) immunoglobulin
- D) Toll-like receptor

77. Which of the following is involved in recognition of intracellular pathogens in innate immune cells:

- A) Toll-like receptors (TLRs)
- B) antibody
- C) NOD-like receptors (NLRs)
- D) complement

78. Which of the following generally does not apply to bone marrow (a primary lymphoid organ) but does apply to secondary lymphoid organs:

- A) cellular proliferation
- B) differentiation of lymphocytes
- C) cellular interaction
- D) antigen-dependent response

79. Mature dendritic cells are capable of which of the following :

A) activating naïve antigen-specific T cells

B) removing red blood cells

C) producing bradykinin

D) extracellular killing of target cells

80. Which of the following applies uniquely with respect to B cells found in secondary lymphoid organs :

A) present as precursor B cells

B) express only IgM

C) terminally differentiate into plasma cells

D) undergo proliferation

81. Which of the following sequence correctly describes lymphocyte migration from lymph nodes to blood?

A) postcapillary venules, efferent lymphatic vessels, thoracic duct, vena cava, heart

B) postcapillary venules, afferent lymphatic vessels, thoracic duct, vena cava, heart

,C) postcapillary venules, efferent lymphatic vessels vena cava, thoracic duct, heart

,D) postcapillary venules, afferent lymphatic vessels vena cava, thoracic duct, heart

82. Cortical cells of the thymus does not include?

a. Hassals corpuscles

b. epithelial cells

c. thymocytes

83. primary lymphoid organs are?

a. thymus

b. bone marrow

c. lymph nodes

d. a and b are correct

84. lymphatics ultimately deliver into?

a. Aorta

b. carotid

c. Alveolar artery

d. Thoracic duct

85. The thymus begins to atrophy and stroma is replaced by adipose fatty tissue?

a. in fetal stage

b. after 40 years of old

c. by 10 years old child

d. by the early teens

86. Neonatal thymectomy results dramatic?

- a. increase in T cells
- b. decrease in T cells**
- c. increase in B cells
- d. decrease in B cells

87. The thymus is responsible for?

- a. Diversity of T cells
- b. self-tolerance of T cells
- c. diversity of NK cells
- d. a and b are correct**

88. clusters of lymphocytes distributed in the lining of the small intestine:

- a. MALT tissue
- b. Peyer's patches**
- c. IgA producing cells
- d. respiratory associated lymphoid tissue

89. The bone marrow stroma cells:

- a. found mainly in yellow bone marrow
- b. contains mesenchymal stem cells**
- c. a and b are correct

90. The area surrounding the small arterioles of the spleen is rich in ?

- a. B and T cells
- b. NK cells
- c. B cells
- d. T cells**

91. The germinal centers found in the cortical region of lymph nodes and the peripheral region of splenic periarteriolar lymphatic tissue :

- A) support the development of immature B and T cells
- B) function in the removal of damaged erythrocytes from the circulation
- C) act as the major source of stem cells and thus help maintain hematopoiesis
- D) provide an infrastructure that on antigenic stimulation contains large populations of B lymphocytes and plasma cells**
- E) are the sites of natural killer T (NKT)-cell differentiation

92. Which of the following IgG subclass cannot pass placenta?

- A. IgG1
- B. IgG2**
- C. IgG3
- D. IgG4

93. Which of the following immunoglobulins can activate complement as a single molecule when bound to an antigen :

- A) IgA
- B) IgE
- C) IgG
- D) IgM**

94. The following is false about the hinge region of IgG? Select one

- a. present in all Ig molecules
- b. lies between CH1 and CH2
- C. structurally constant for all Ig**
- d. rich in pro and Cys amino acids
- e. confers flexibility to the Fab region

95. The following sentence is false about the breast mother milk in comparison to artificial powdered milk? Select one:

- a. has higher Phagocytic activity in the intestinal tissue of infants
- b. has higher NK cells activity in the intestinal tissue of infants
- c. has higher antibodies
- d. has higher concentration of vitamins**
- e. shows more efficient Opsonization

96. Functional properties of immunoglobulins such as binding to Fc receptors are associated with :

- A) light chains
- B) J chains
- C) disulfide bonds
- D) heavy chains**

97. The idiotype of an antibody molecule is determined by the amino acid sequence of the:

- A) constant region of the light chain
- B) variable region of the light chain
- C) constant region of the heavy chain
- D) constant regions of the heavy and light chains
- E) variable regions of the heavy and light chain**

98. Which of the following statements is TRUE of IgD? Select one:

- a. IgD is associated with allergy
- b. IgD is resistant to proteolytic degradation
- c. IgD is abundant in human milk
- d. IgD is present mainly as a surface receptor on B-cells**

99. Which of the following statements is TRUE about IgM? Select one:

- a. IgM is the main class of the isohemagglutinins of anti A and anti B antigens**
- b. IgM is usually of high intrinsic affinity
- c. IgM is most commonly tetrameric

100. Primary and secondary antibody responses differ in:

- A) the predominant isotype generated
- B) the number of lymphocytes responding to antigen
- C) the time it takes for measurable amounts of antibodies to appear in the serum
- D) the biologic functions manifested by the Ig isotypes produced
- E) all of the above

101. Which of the following statement does not apply to IgG?

- a. Neutralizes bacterial toxins
- b. are not synthesized by the fetus
- c. can fix complement
- d. appears earlier than IgM in the primary immune response

102. If an individual who is highly allergic to cat dander is exposed to a pet cat in a friend's house, which class of immunoglobulin would most likely be found to be elevated soon after this exposure :

- A) IgA
- B) IgE
- C) IgG
- D) IgM
- E) IgD

